

GOODSOL LEE

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RESEARCH INTEREST

High-Quality Real-Time Communications (e.g., VR and AR) over Wireless Networks

- Wireless Networks, WebRTC, Video Analytics, Networking for AI, and Congestion Control

EDUCATION

Seoul National University

Seoul, South Korea

Ph.D. in Electrical and Computer Engineering

Sep 2018 – Feb 2026

- Advisor: Prof. Kyunghan Lee and Prof. Saewoong Bahk

- Thesis: Towards Seamless Mobile Real-Time Communications with QoE-Driven RAN

Seoul National University

Seoul, South Korea

B.S. in Electrical and Computer Engineering

Mar 2012 – Aug 2018

EXPERIENCES

Nokia Bell Labs

New Providence, New Jersey, USA

Network System Researcher

May 2026 – Present

Microsoft Research Asia

Beijing, China

Research Intern, Host: Dr. Juheon Yi

Nov 2025 – Feb 2026

University of Colorado Boulder

Boulder, CO, USA

Visiting Scholar, Host: Prof. Sangtae Ha

June 2024 – Aug 2025

Purdue University

West Lafayette, IN, USA

Visiting Scholar, Host: Prof. Kwang Taik Kim & Prof. Mung Chiang

Dec 2023 – May 2024

Samsung System LSI

Hwaseong, Korea

Software Engineering Intern

Summer 2023

Seoul National University

Seoul, South Korea

Research Assistant, Advisor: Prof. Kyunghan Lee and Prof. Saewoong Bahk

Sep 2018 – Feb 2026

SK Hynix

Seongnam, Korea

Undergraduate Intern

Summer 2017

SELECTED PUBLICATIONS

Conference

1. **Goodsol Lee**, Junhong Min, Seyeon Kim, Juheon Yi, Kwang Taik Kim, Mung Chiang, Sangtae Ha, Kyunghan Lee, and Saewoong Bahk, "QCON: Seamless QoE-Aware 5G Streaming via Multi-Connectivity," in *23rd USENIX Symposium on Networked Systems Design and Implementation (NSDI '26)*, USENIX, 2026
2. **Goodsol Lee**, Seyeon Kim, Juheon Yi, Junhong Min, Sangtae Ha, Kyunghan Lee, and Saewoong Bahk, "PAVE: Mitigating Non-Congestive Delay for Seamless Video Calls over NextG Mobile Networks," in *IEEE INFOCOM 2026-IEEE Conference on Computer Communications*, IEEE, 2026
3. Juheon Yi, MinKyung Jeong, Seokgyeong Shin, **Goodsol Lee**, Daehyeok Kim, and Youngki Lee, "Pendulum: Network-Compute Joint Scheduling for Efficient and Accurate MEC Live Video Analytics," in *IEEE INFOCOM 2026-IEEE Conference on Computer Communications*, IEEE, 2026
4. Juheon Yi, **Goodsol Lee**, Seokgyeong Shin, MinKyung Jeong, Daehyeok Kim, and Youngki Lee, "Towards End-to-End Latency Guarantee in MEC Live Video Analytics with App-RAN Mutual Awareness," in *23nd ACM International Conference on Mobile Systems, Applications, and Services (MobiSys)*, ACM, 2025
5. Juhun Shin, **Goodsol Lee**, Jeongyeop Paek, and Saewoong Bahk, "César: Cellular Resource Scheduling-Aware Congestion Control," in *IEEE INFOCOM 2025-IEEE Conference on Computer Communications*, IEEE, 2025

6. **Goodsol Lee**, Siyoung Choi, Junseok Kim, Youngseok Kim, and Saewoong Bahk, "Prosch: Proxy Aided Secondary Cell Handover in Ultra-Dense Mmwave Network," in *2020 IEEE Wireless Communications and Networking Conference (WCNC)*, IEEE, 2020, pp. 1–6

Journal

1. Junseok Kim, **Goodsol Lee**, Seongwon Kim, Tarik Taleb, Sunghyun Choi, and Saewoong Bahk, "Two-Step Random Access for 5G System: Latest Trends and Challenges," *IEEE Network*, vol. 35, no. 1, pp. 273–279, 2020

TECHNICAL SKILLS

- **Programming Languages:** C/C++, Java, Python
- **Network Systems:** NS-3 Simulator, srsRAN, Openairinterface5G, XCAL Cellular Analyzer, WebRTC, TCP, QUIC
- **Operating Systems:** Android, Linux
- **ML Frameworks:** Llama.cpp, Tensorflow, PyTorch
- **Languages:** Korean (native), English (fluent)

AWARDS

- Gold Prize (1st prize in Communications & Network division) from Samsung HumanTech Paper Awards, Feb. 2026. (As the first author)
- Silver Prize (2nd prize in Communications & Network division) from Samsung HumanTech Paper Awards, Feb. 2025. (As a co-author)

PROFESSIONAL SERVICES

Review

- **Conferences:** WNS3 2021, IEEE Globecom 2020
- **Journals:** IEEE Transactions on Networking (2026), IEEE Transactions on Mobile Computing (2025), IEEE Transactions on Vehicular Technology (2025), IEEE Transactions on Network Science and Engineering (2025), Computer Networks (2025), IEEE Network Letters (2024)

TPC Member

- **Conferences:** IEEE INFOCOM 2027, MobiSys 2026 Workshop on NetAISys

External Review

- **Conferences:** ACM MobiCom 2026, IEEE INFOCOM 2025, ACM MobiCom 2025, ACM MobiHoc 2024, IEEE INFOCOM 2024, WiOpt 2022, ACM MobiHoc 2021
- **Journals:** IEEE Transactions on Networking, Journal of Communications and Networks

Invited Talks

"Towards Seamless Mobile Real-Time Communications with QoE-Driven RAN"

-Nokia Bell Labs

Oct, 2025

"OpenAirInterface5G Tutorial: From Basics to Publication"

-Korea University

Oct, 2025